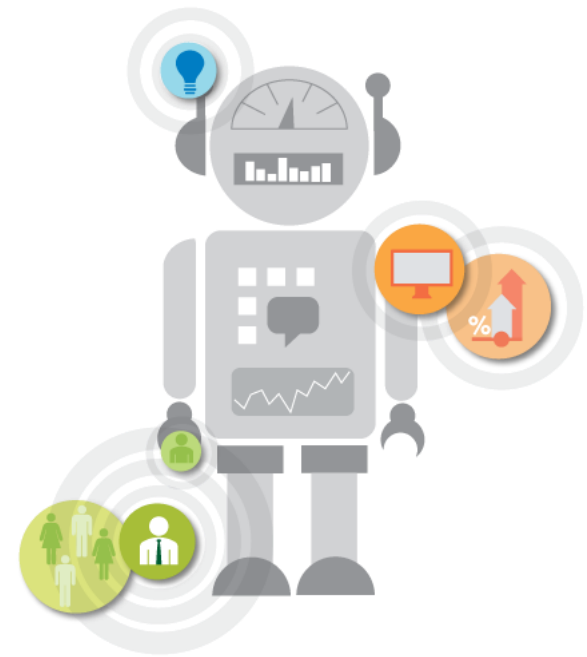


AI and Society

CSET-485

Dr. Pulkit Dwivedi



UNIT - 1



Regression: Simple Linear Regression, Multiple Linear Regression and Logistic Regression

Regression Analysis

- Regression analysis is a statistical method to model the relationship between a dependent (target) and independent (predictor) variables with one or more independent variables.
- It helps us to understand how the value of the dependent variable is changing corresponding to an independent variable when other independent variables are held fixed.
- It predicts continuous/real values such as temperature, age, salary, price, etc.
- It is a supervised learning technique which helps in finding the correlation between variables and enables us to predict the continuous output variable based on the one or more predictor variables.
- It is mainly used for prediction, forecasting, time series modeling, and determining the causal-effect relationship between variables.

Regression Analysis

- **Example:** Suppose there is a marketing company A, who does various advertisement every year and get sales on that. The below list shows the advertisement made by the company in the last 5 years and the corresponding sales:

Advertisement	Sales
\$90	\$1000
\$120	\$1300
\$150	\$1800
\$100	\$1200
\$130	\$1380
\$200	??

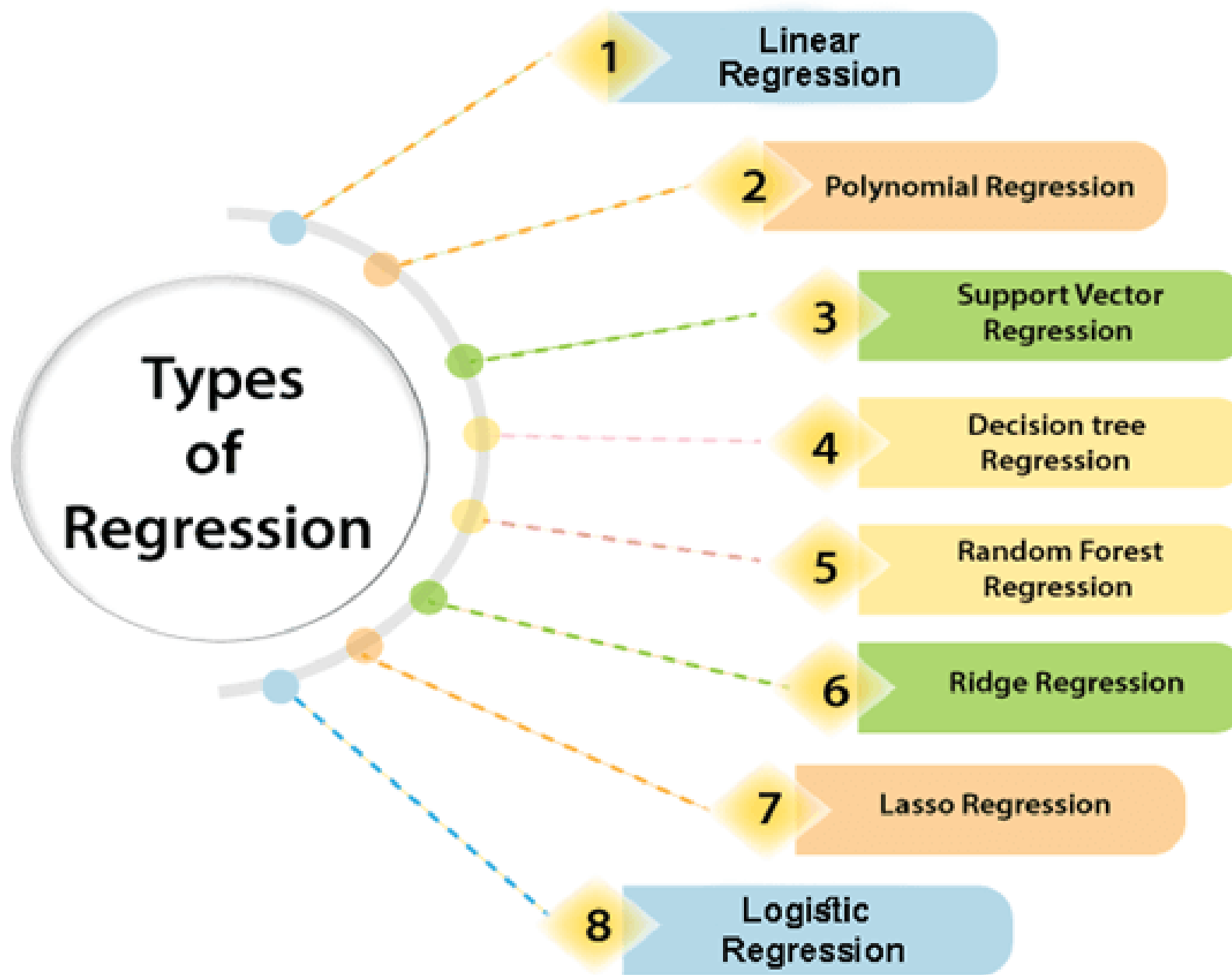
- Now, the company wants to do the advertisement of \$200 in the year 2019 and wants to know the prediction about the sales for this year. So to solve such type of prediction problems in machine learning, we need regression analysis.

Regression Analysis

- In Regression, we plot a graph between the variables which best fits the given datapoints, using this plot, the machine learning model can make predictions about the data.
- In simple words, **"Regression shows a line or curve that passes through all the datapoints on target-predictor graph in such a way that the vertical distance between the datapoints and the regression line is minimum."**
- The distance between datapoints and line tells whether a model has captured a strong relationship or not.

Regression Analysis

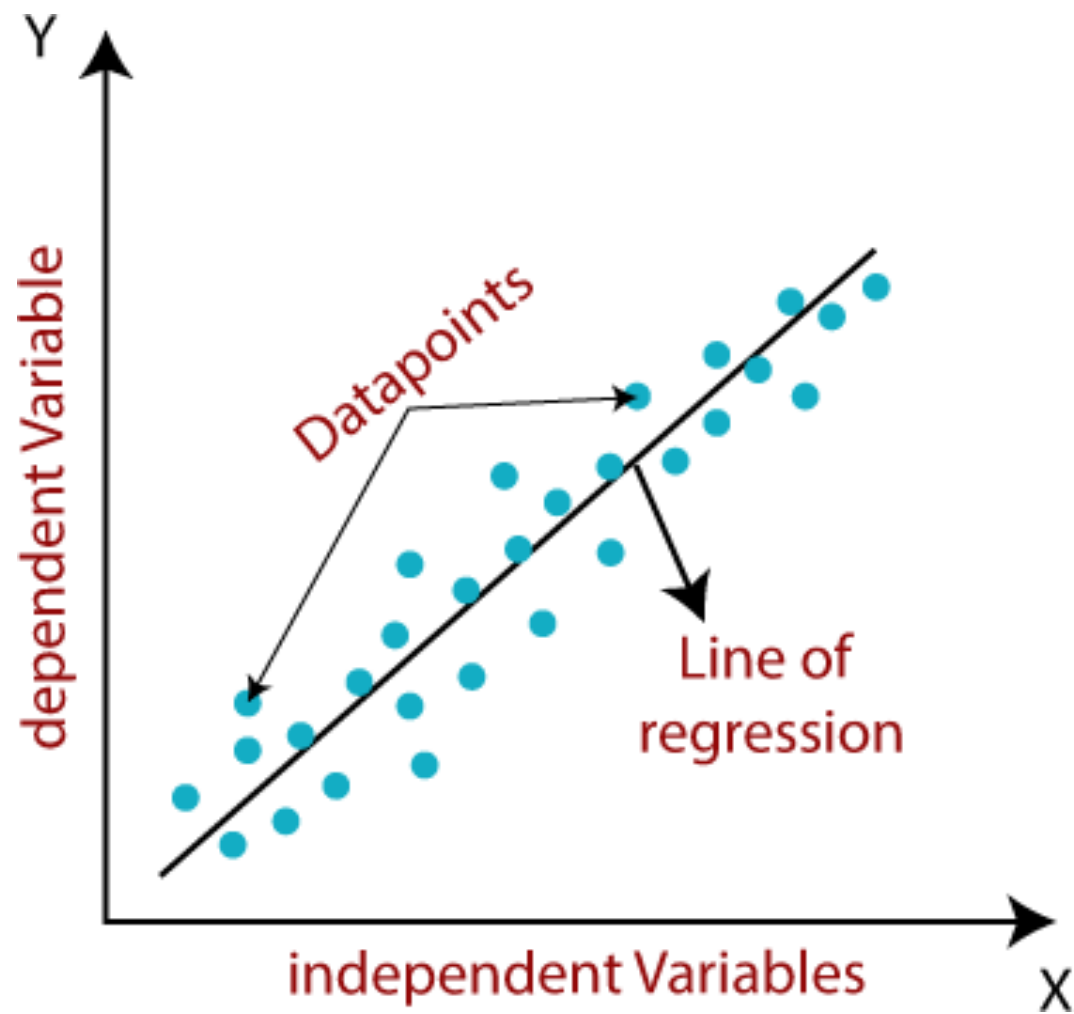
- **Dependent Variable:** The main factor in Regression analysis which we want to predict or understand is called the dependent variable. It is also called target variable.
- **Independent Variable:** The factors which affect the dependent variables or which are used to predict the values of the dependent variables are called independent variable, also called as a predictor.
- **Multicollinearity:** If the independent variables are highly correlated with each other than other variables, then such condition is called Multicollinearity. It should not be present in the dataset, because it creates problem while ranking the most affecting variable.



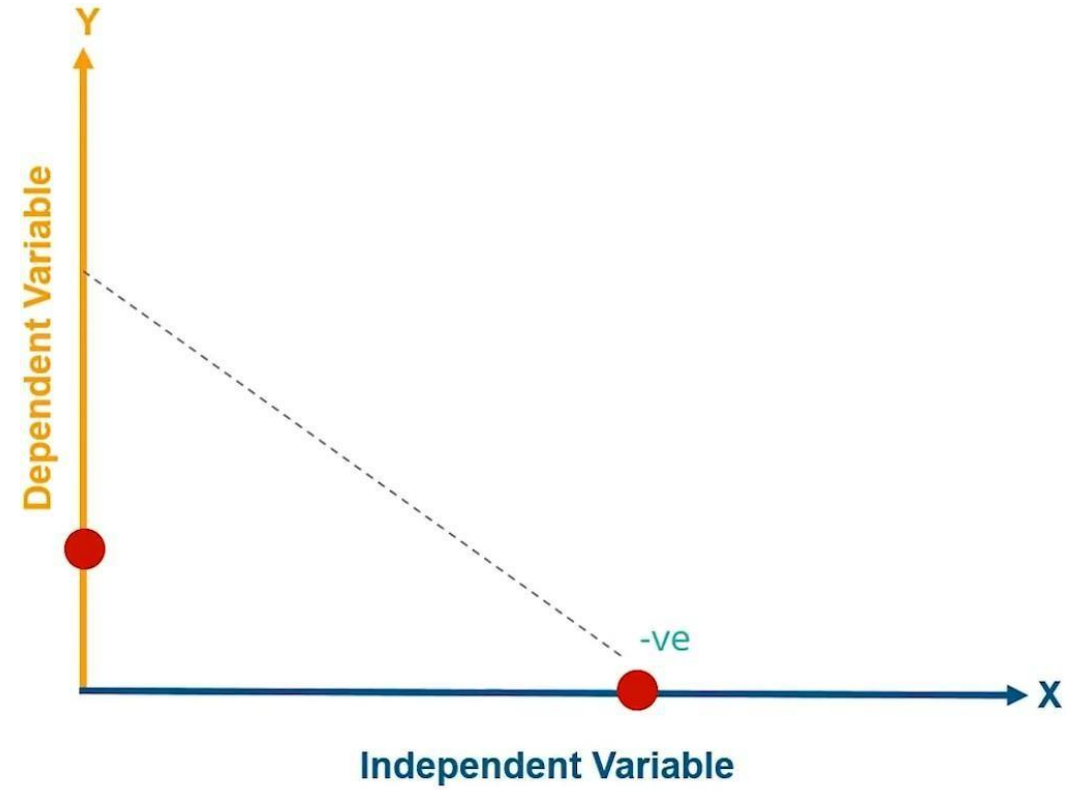
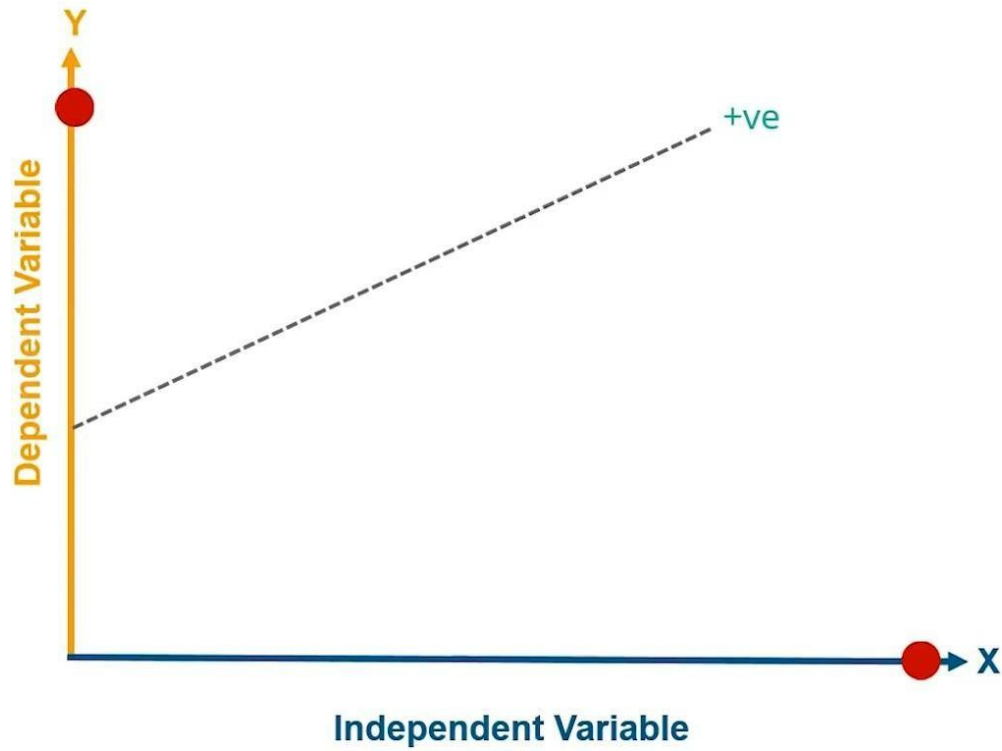
Linear Regression

- Linear regression is one of the easiest and most popular Machine Learning algorithms.
- Linear regression algorithm shows a linear relationship between a dependent (X) and one or more independent (y) variables, hence called as linear regression.
- Since linear regression shows the linear relationship, which means it finds how the value of the dependent variable is changing according to the value of the independent variable.
- The linear regression model provides a sloped straight line representing the relationship between the variables.

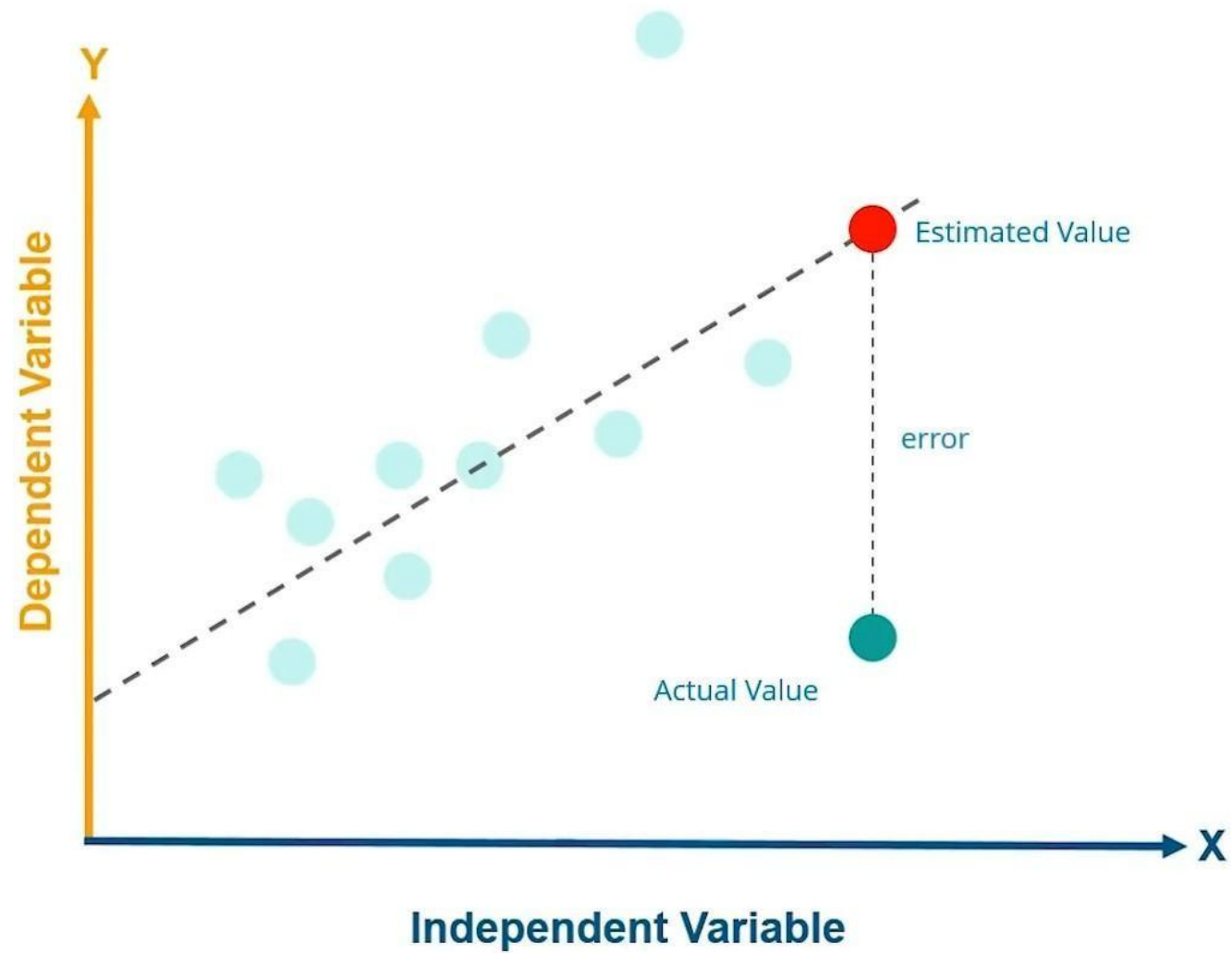
Linear Regression



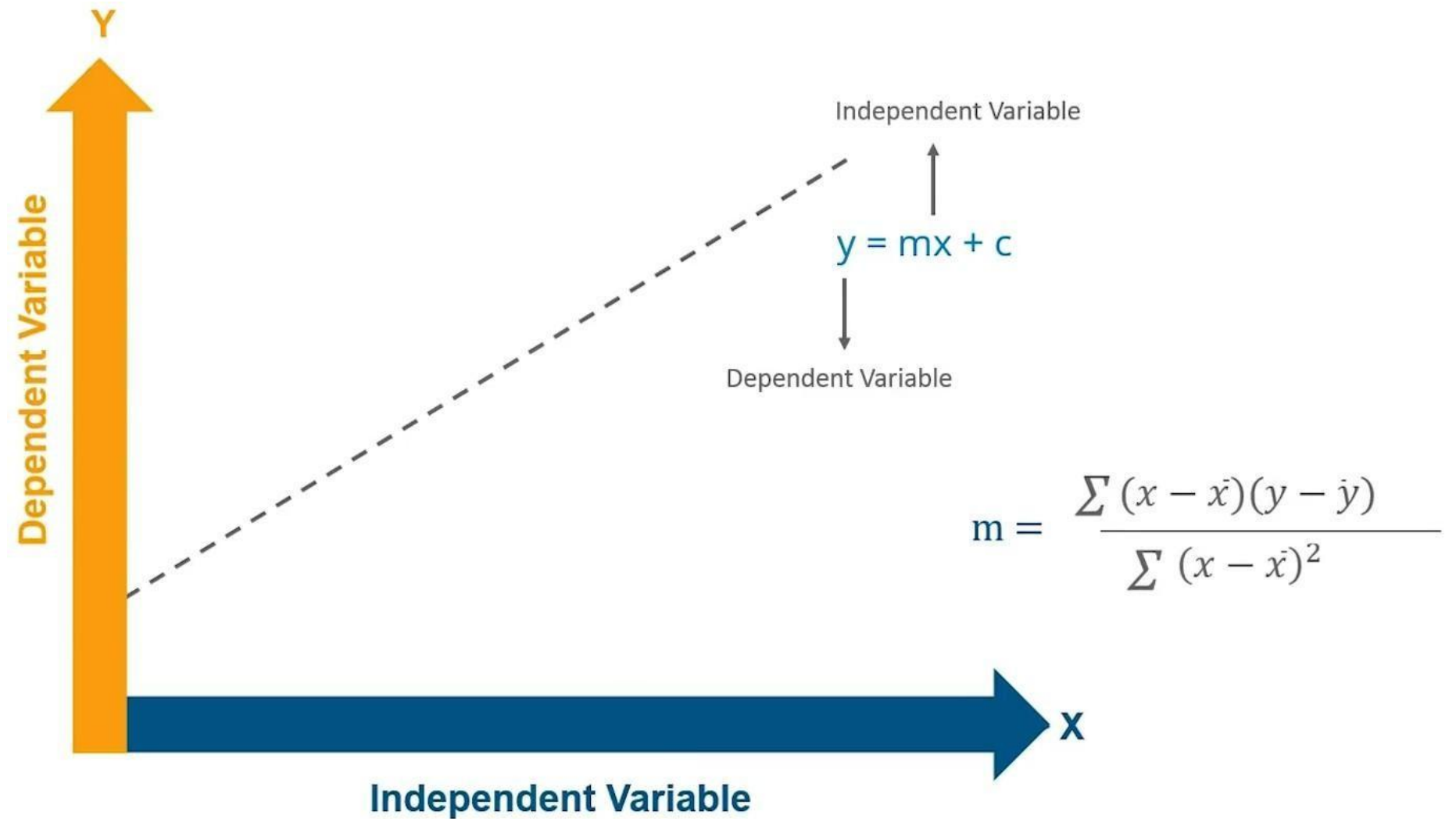
Linear Regression



Linear Regression



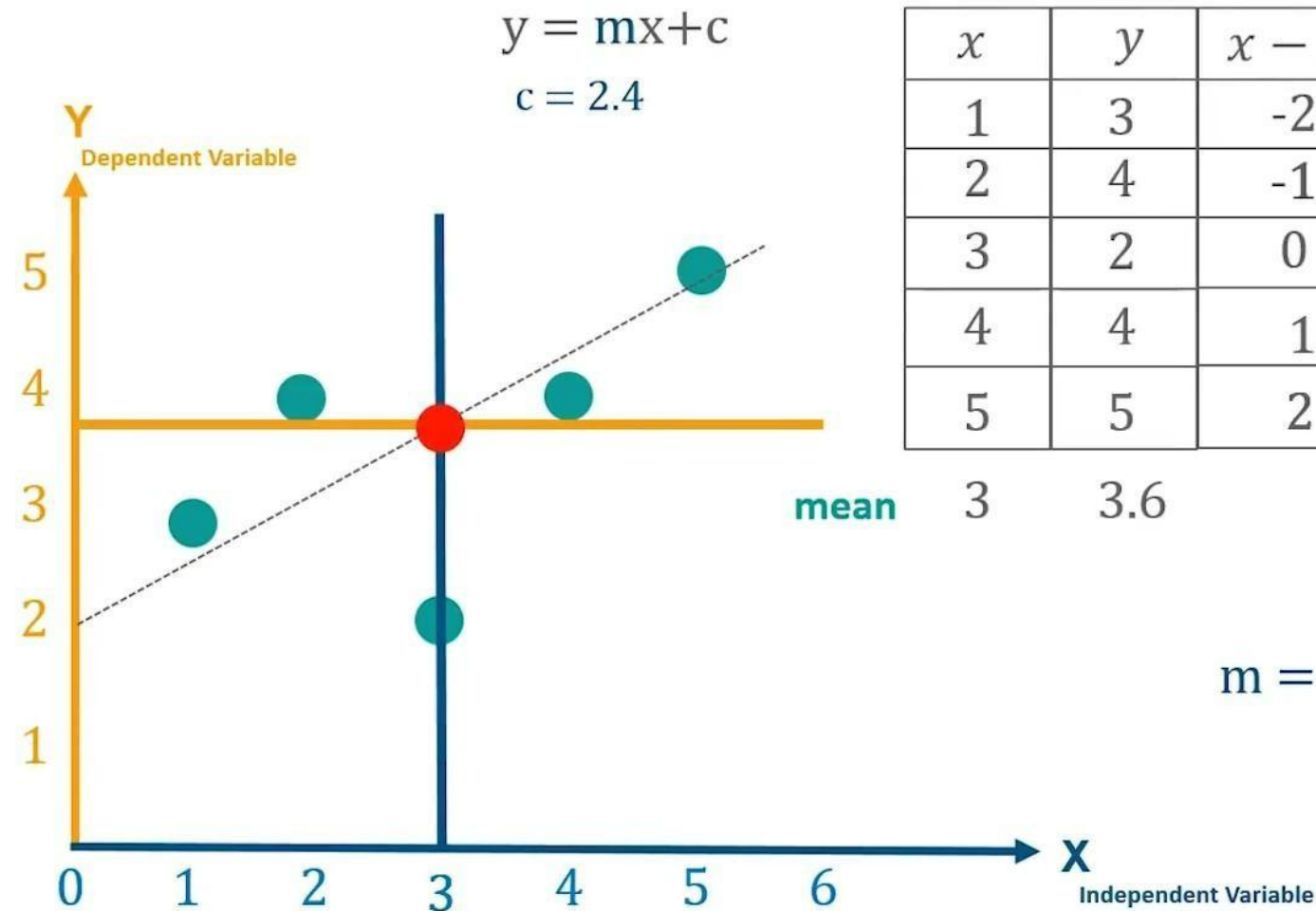
Linear Regression



Linear Regression

x	y
1	3
2	4
3	2
4	4
5	5

Linear Regression



x	y	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})^2$	$(x - \bar{x})(y - \bar{y})$
1	3	-2	-0.6	4	1.2
2	4	-1	0.4	1	-0.4
3	2	0	-1.6	0	0
4	4	1	0.4	1	0.4
5	5	2	1.4	4	2.8
3	3.6			$\Sigma = 10$	$\Sigma = 4$

$$m = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{4}{10}$$

$$\begin{aligned} m &= 0.4 \\ c &= 2.4 \\ y &= 0.4x + 2.4 \end{aligned}$$

R-squared or Coefficient of Determination

Sum of Squares Due to Regression

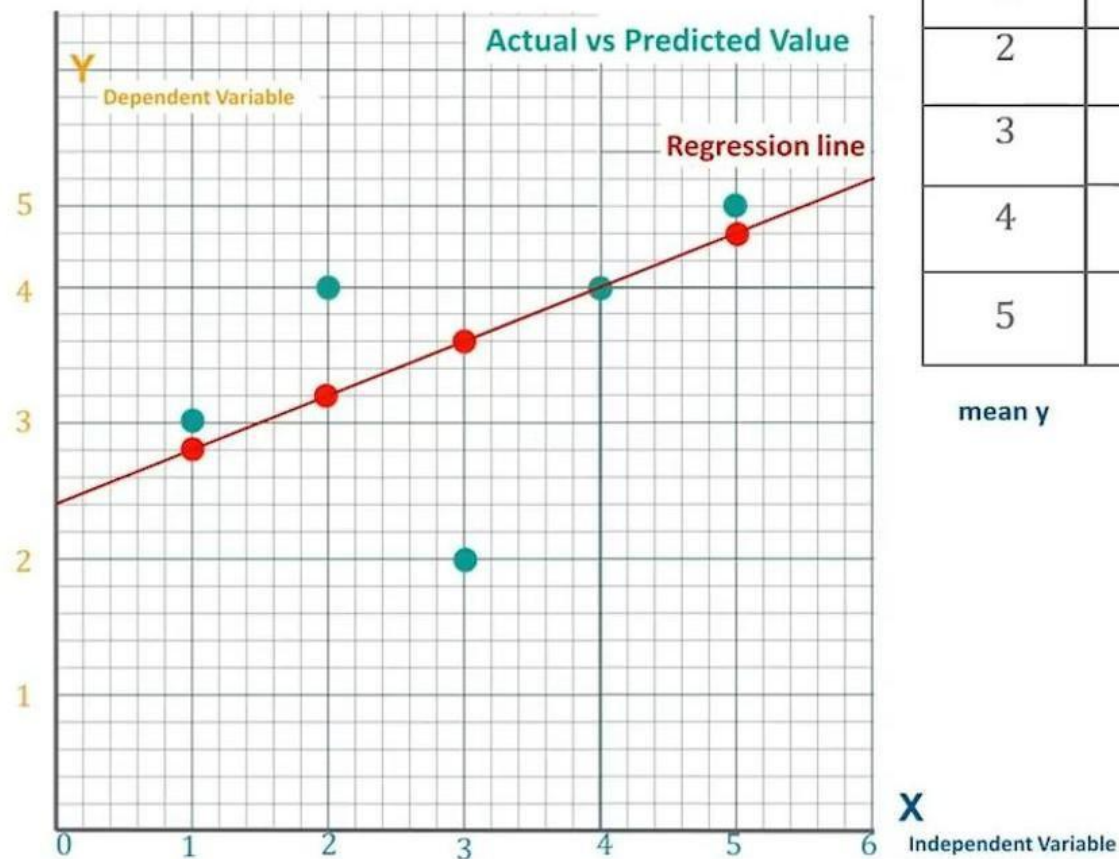
$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

$$R^2 = SSR/SST$$

Total Sum of Squares

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

Calculation of R-squared



x	y	$y - \bar{y}$	$(y - \bar{y})^2$	y_p	$(y_p - \bar{y})$	$(y_p - \bar{y})^2$
1	3	-0.6	0.36	2.8	-0.8	0.64
2	4	0.4	0.16	3.2	-0.4	0.16
3	2	-1.6	2.56	3.6	0	0
4	4	0.4	0.16	4.0	0.4	0.16
5	5	1.4	1.96	4.4	0.8	0.64
mean y		3.6	Σ 5.2		Σ 1.6	

$$R^2 = \frac{1.6}{5.2} = \frac{\Sigma (y_p - \bar{y})^2}{\Sigma (y - \bar{y})^2} = 0.30$$

Error Terms: Mean absolute error (MAE)

- Mean absolute error, also known as L1 loss is one of the simplest loss functions and an easy-to-understand evaluation metric.
- It is calculated by taking the absolute difference between the predicted values and the actual values and averaging it across the dataset. .
- The lower the MAE, the higher the accuracy of a model.

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Error Terms: Mean Squared Error (MSE)

- MSE is one of the most common regression loss functions.
- In Mean Squared Error also known as L2 loss, we calculate the error by squaring the difference between the predicted value and actual value and averaging it across the dataset.
- MSE will never be negative since the errors are squared.
- The value of the error ranges from zero to infinity.
- MSE increases exponentially with an increase in error. A good model will have an MSE value closer to zero.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Error Terms: Root Mean Squared Error (RMSE)

- RMSE is computed by taking the square root of MSE.
- RMSE value with zero indicates that the model has a perfect fit.
- The lower the RMSE, the better the model and its predictions. A higher RMSE indicates that there is a large deviation from the residual to the ground truth.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

Multiple Linear Regression



Multiple Linear Regression

Simple Regression :

$$y = b_0 + b_1 x$$

Only one Dependent
Only one Independent

Multiple Linear Regression :

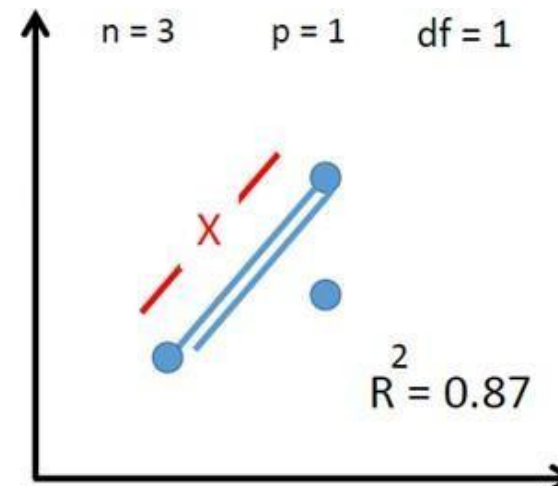
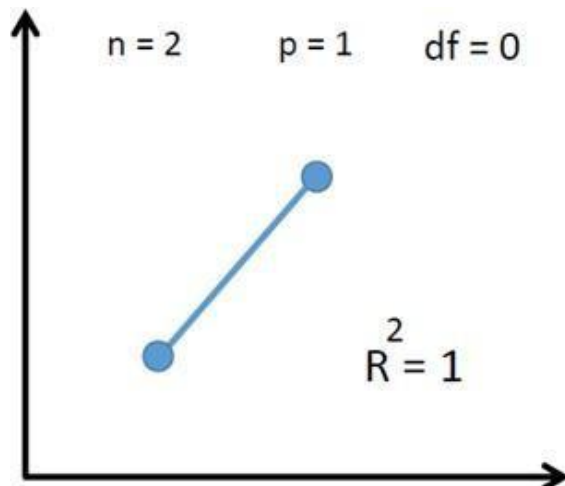
$$y = b_0 + b_1 x_1 + b_2 x_2 + \cdots + b_n x_n$$

Degrees of Freedom

- Degrees of freedom refers to the maximum number of logically independent values, which are values that have the freedom to vary, in the data sample.

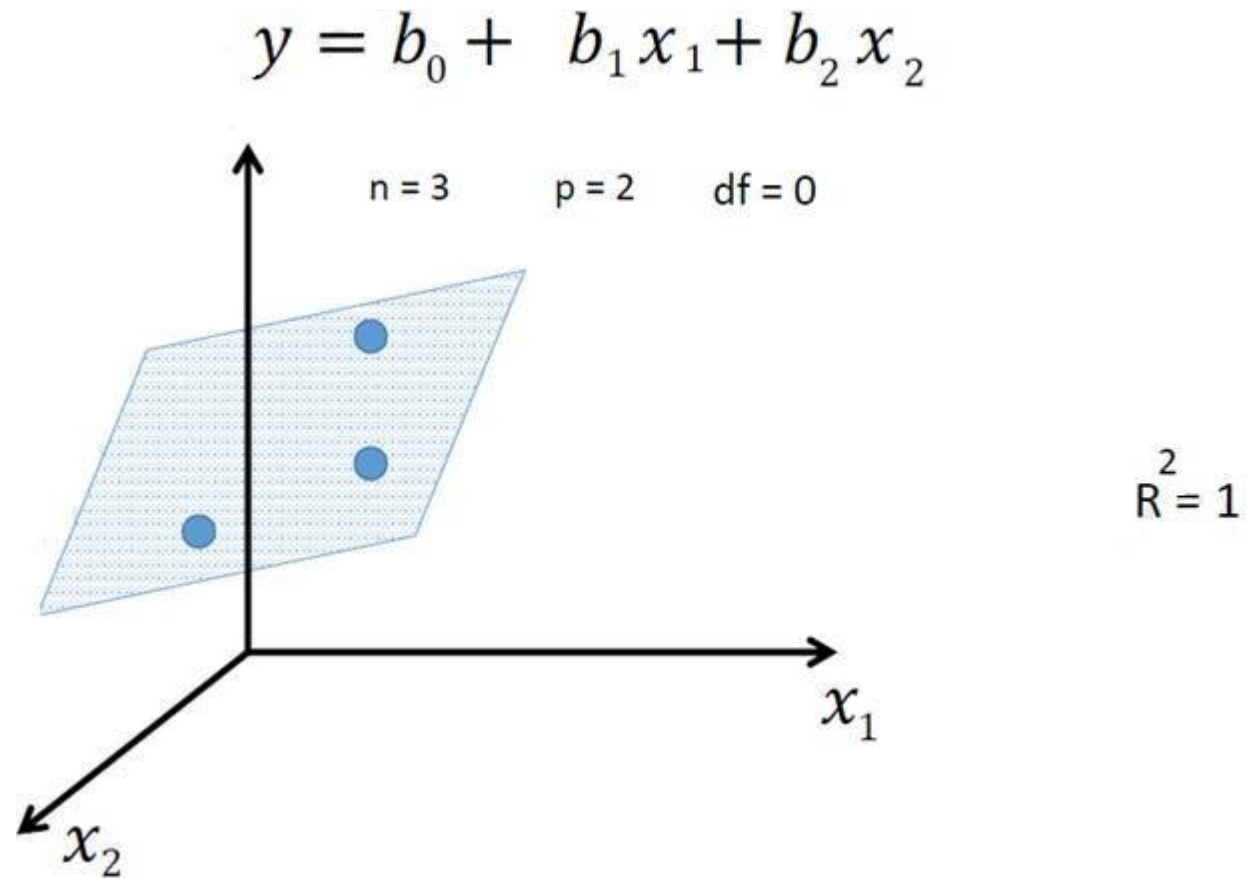
$$df = n - p - 1$$

Number of Observations Number of variables



$$y = b_0 + b_1 x_1$$

Degrees of Freedom



Degrees of Freedom

- Adding more variables, increases the value of R-squared.
- Higher the value of R-squared, variation in Y is better explained by variation in X.
- Hence, add more variables..!!

Degrees of Freedom

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- Hence, add more variables..!!

• **NO...!!!!!!**

Adjusted R - squared

- Adjusted R-Squared punishes addition of non-important variables.

$$\bar{R}^2 = 1 - \frac{(1 - R^2) * (n - 1)}{n - p - 1}$$

R = sampled R-squared

p = Number of independent variables

n = sample size or number of observations

Adjusted R - squared

- When we increase number of variables, R-squared increases

N	p	R-Squared
50	10	0.80
50	12	0.82
50	15	0.83
50	20	0.84

Adjusted R - squared

- When we increase number of variables, R-squared increases but adjusted R-squared does not increase.
- When $p = 20$, it decreases to 0.73.

N	p	R-Squared	Adjusted R-Squared
50	10	0.80	0.75
50	12	0.82	0.76
50	15	0.83	0.75
50	20	0.84	0.73

Advantages of Linear Regression

- **Simplicity:** Linear regression is a simple and straightforward technique that is easy to understand and interpret.
- **Prediction:** Linear regression is a useful tool for predicting future values of the dependent variable based on the values of the independent variable(s). It can be used to forecast trends, identify patterns, and estimate future outcomes.
- **Flexibility:** Linear regression can be used with both continuous and categorical independent variables, making it a versatile technique for a wide range of applications.
- **Efficiency:** Linear regression is computationally efficient, making it suitable for handling large datasets with a large number of observations and variables.
- **Model selection:** Linear regression can be used as a starting point for more complex modeling techniques, such as polynomial regression, logistic regression, and more, to help identify which variables are important in predicting the outcome.

Disadvantages of Linear Regression

- **Linearity assumption:** Linear regression assumes that the relationship between the dependent variable and the independent variable(s) is linear. If the relationship is nonlinear, the results of the model may be misleading.
- **Outliers:** Linear regression is sensitive to outliers, which can have a disproportionate influence on the regression line and affect the accuracy of the predictions.
- **Multicollinearity:** If the independent variables in a linear regression model are highly correlated, it can be difficult to determine their individual effects on the dependent variable. This is known as multicollinearity, and it can lead to unreliable estimates of the coefficients.
- **Overfitting:** If a linear regression model is too complex or includes too many independent variables, it may overfit the data, meaning that it fits the noise in the data as well as the signal, and may not generalize well to new data.

Applications of Linear Regression: Advertising

- Businesses often use linear regression to understand the relationship between advertising spending and revenue.

$$\text{revenue} = \beta_0 + \beta_1(\text{ad spending})$$

- The coefficient β_0 would represent total expected revenue when ad spending is zero.
- If β_1 is negative, it would mean that more ad spending is associated with less revenue.
- If β_1 is close to zero, it would mean that ad spending has little effect on revenue.
- And if β_1 is positive, it would mean more ad spending is associated with more revenue.
- Depending on the value of β_1 , a company may decide to either decrease or increase their ad spending.

Applications of Linear Regression: Medical

- Medical researchers often use linear regression to understand the relationship between drug dosage and blood pressure of patients.

$$\text{blood pressure} = \beta_0 + \beta_1(\text{dosage})$$

- β_0 represent the expected blood pressure when dosage is zero and β_1 represent the average change in blood pressure when dosage is increased by one unit.
- If β_1 is negative, it would mean that an increase in dosage is associated with a decrease in BP
- If β_1 is close to zero, it would mean that an increase in dosage is associated with no change in BP
- If β_1 is positive, it would mean that an increase in dosage is associated with an increase in BP
- Depending on the value of β_1 , researchers may decide to change the dosage given to a patient.

Applications of Linear Regression: Agriculture

- Agricultural scientists often use linear regression to measure the effect of fertilizer and water on crop yields.

$$\text{crop yield} = \beta_0 + \beta_1(\text{amount of fertilizer}) + \beta_2(\text{amount of water})$$

- The coefficient β_0 would represent the expected crop yield with no fertilizer or water.
- The coefficient β_1 would represent the average change in crop yield when fertilizer is increased by one unit, assuming the amount of water remains unchanged.
- The coefficient β_2 would represent the average change in crop yield when water is increased by one unit, assuming the amount of fertilizer remains unchanged.
- Depending on the values of β_1 and β_2 , the scientists may change the amount of fertilizer and water used to maximize the crop yield.

Applications of Linear Regression: Sports

- Data scientists for professional sports teams often use linear regression to measure the effect that different training regimens have on player performance.

$$\text{points scored} = \beta_0 + \beta_1(\text{yoga sessions}) + \beta_2(\text{weightlifting sessions})$$

- The coefficient β_0 would represent the expected points scored for a player who participates in zero yoga sessions and zero weightlifting sessions.
- The coefficient β_1 would represent the average change in points scored when weekly yoga sessions is increased by one, assuming the number of weekly weightlifting sessions remains unchanged.
- The coefficient β_2 would represent the average change in points scored when weekly weightlifting sessions is increased by one, assuming the number of weekly yoga sessions remains unchanged.
- Depending on the values of β_1 and β_2 , the data scientists may recommend that a player participates in more or less weekly yoga and weightlifting sessions in order to maximize their points scored.